

A randomized controlled trial on the anti-inflammatory effects of curcumin in patients with chronic sulphur mustard-induced cutaneous complications.



BACKGROUND: Chronic cutaneous complications such as pruritus are among the very frequent complaints of sulphur mustard (SM)-exposed patients. The present trial investigated the impact of curcumin on serum inflammatory biomarkers and their association with pruritus severity and quality of life (QoL). **METHODS:** This was a randomized, double-blind trial among 96 male Iranian veterans (age 37-59 y) who were suffering from chronic SM-induced pruritic skin lesions. Patients were randomly assigned to curcumin (1 g/d, n = 46) or placebo (n = 50) for four weeks. Serum concentrations of interleukins 6 (IL-6) and 8 (IL-8) together with high-sensitivity C-reactive protein (hs-CRP) and calcitonin gene-related peptide (CGRP) were measured at baseline and at the end of the trial. Assessment of pruritus severity was performed using the pruritus score and QoL using the Dermatology Life Quality Index (DLQI). **RESULTS:** Serum IL-8 and hs-CRP were significantly reduced in both groups but the magnitude of reduction was greater in the curcumin group ($P < 0.001$). Serum CGRP was only decreased in the curcumin group ($P < 0.001$). No significant change was observed in serum IL-6. There were significant correlations between CGRP and IL-6 changes ($P = 0.011$) and between DLQI and IL-8 changes ($P = 0.026$) in the curcumin group. In the curcumin group, changes in serum IL-8 concentrations were found as the significant predictor of DLQI scores ($P = 0.026$) but none of the independent variables could predict pruritus scores. **CONCLUSIONS:** Curcumin supplementation effectively mitigates inflammation in patients suffering from chronic SM-induced cutaneous complications. This anti-inflammatory effect might account for the observed pruritus alleviation and QoL improvement by this phytochemical.